

Clustering in Unsupervised Learning

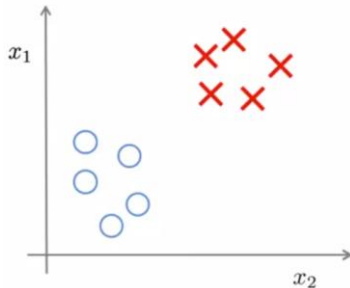
Chapter 9: Unsupervised Learning Techniques
Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Gurelien Geron, 3e, O'Reilly Media, 2022.

Unsupervised Learning

- An unsupervised learning algorithm finds structures in data.

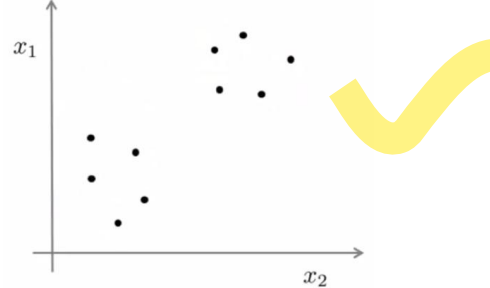
Supervised Learning

- Labeled data
- Training set:
 $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})\}$



Unsupervised Learning

- Unlabeled data
 - Training set:
 $\{x^{(1)}, x^{(2)}, \dots, x^{(n)}\}$
- all points similar since no labels



- There are various types of unsupervised learning including clustering, anomaly detection, dimensionality reduction, and association rule learning.

Clustering

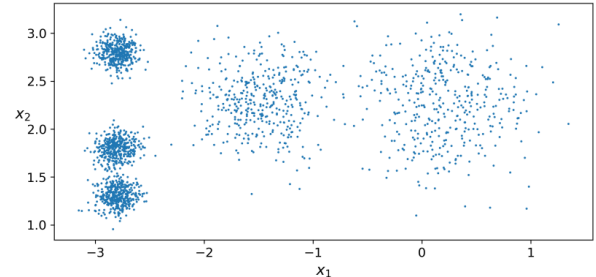
- Clustering is an unsupervised learning technique used to group similar data points into clusters.
- Unlike supervised learning, clustering does not use labeled data.
- The goal is to ensure that data points within the same cluster are more similar to each other than to points in other clusters.
- Clustering is used in a wide variety of applications, including:
 - Customer / market segmentation: Group customers according to certain choices, purchases, etc. This is useful for marketing campaigns and recommendation systems.
 - Social network analysis: Identify group of friends in a social network.
 - Data analysis while preprocessing data: When analyzing a new dataset, it is often useful to first discover clusters of similar instances, as it is often easier to analyze clusters separately.
 - Semi-supervised learning: If you only have a few labels, you could perform clustering and propagate the labels to all the instances in the same cluster.
 - Search engines: Putting similar content in a cluster simplifies search process.
 - Image Segmentation: Club similar pixels in the image together. By clustering pixels according to their color, then replacing each pixel's color with the mean color of its cluster, it is possible to reduce the number of different colors in the image considerably. This technique is used in many object detection and tracking systems, as it makes it easier to detect the contour of each object.
- Some popular clustering algorithms are:
 - K-Means Clustering
 - Hierarchical Clustering
 - DBSCAN

K-Means Clustering Algorithm

- K-Means was proposed by Stuart Lloyd at Bell Labs in 1957 as a technique for pulse-code modulation. It was later published outside the company in 1982 in the paper “*Least Squares Quantization in PCM*”.
- K-Means is a centroid-based clustering algorithm that partitions data into K clusters.
- Each cluster is represented by a centroid, which is the mean of the data points in that cluster.
- The algorithm iteratively assigns data points to the nearest centroid and updates centroids until convergence.
- The objective is to minimize the within-cluster variance (inertia).

Example

We try to find the center of each blob, called a centroid, and assign each instance to the nearest blob.



Algorithm

1. Choose the number of clusters K.
 2. Select K random points, preferably from the data, as centroids.
 3. Assign all the points to the closest cluster centroid.
 4. Re-compute the centroids of newly formed clusters.
 5. Repeat steps 3 and 4 until the stopping condition is met.
- Some possible stopping criteria are as follows:
 - Centroids do not change significantly.
 - Data points remain in the same clusters.
 - Maximum number of iterations is reached.
 - Hence K-means clustering algorithm iteratively performs two steps:
 - **Cluster-assignment step:** Assign each training example $x^{(i)}$ to the cluster with the nearest centroid.
$$c^{(i)} = \min_k ||x^{(i)} - \mu_k||$$

Where $c^{(i)}$ is the cluster assigned to example i , and μ_k is the centroid of cluster K .
Here both $x^{(i)}$ and μ_k are m dimensional vectors with m being the number of features.
 - **Move-centroid step:** Update each centroid by computing the mean of all points assigned to that cluster.
$$\mu_k = \text{average (mean) of points assigned to cluster } K.$$
 - If a cluster centroid has no points assigned to it, then:
 - randomly reinitialize the centroid.
 - remove (drop) the centroid from the clustering process.
 - K-Means clustering can also be applied to non-separable data, although clusters may overlap.

Input:

- K = number of clusters
- Training set: $\{x^{(1)}, x^{(2)}, x^{(3)}, \dots, x^{(n)}\}$, where $x^{(i)}$ is an m -dimensional vector.

Randomly initialize K centroids: $\mu_1, \mu_2, \dots, \mu_K$

Repeat until converge:

```
for i = 1 to n
     $c^{(i)} = \min_k ||x^{(i)} - \mu_k||$ 
for k = 1 to K
     $\mu^{(k)} = \text{average (mean) of points assigned to cluster } k$ 
```

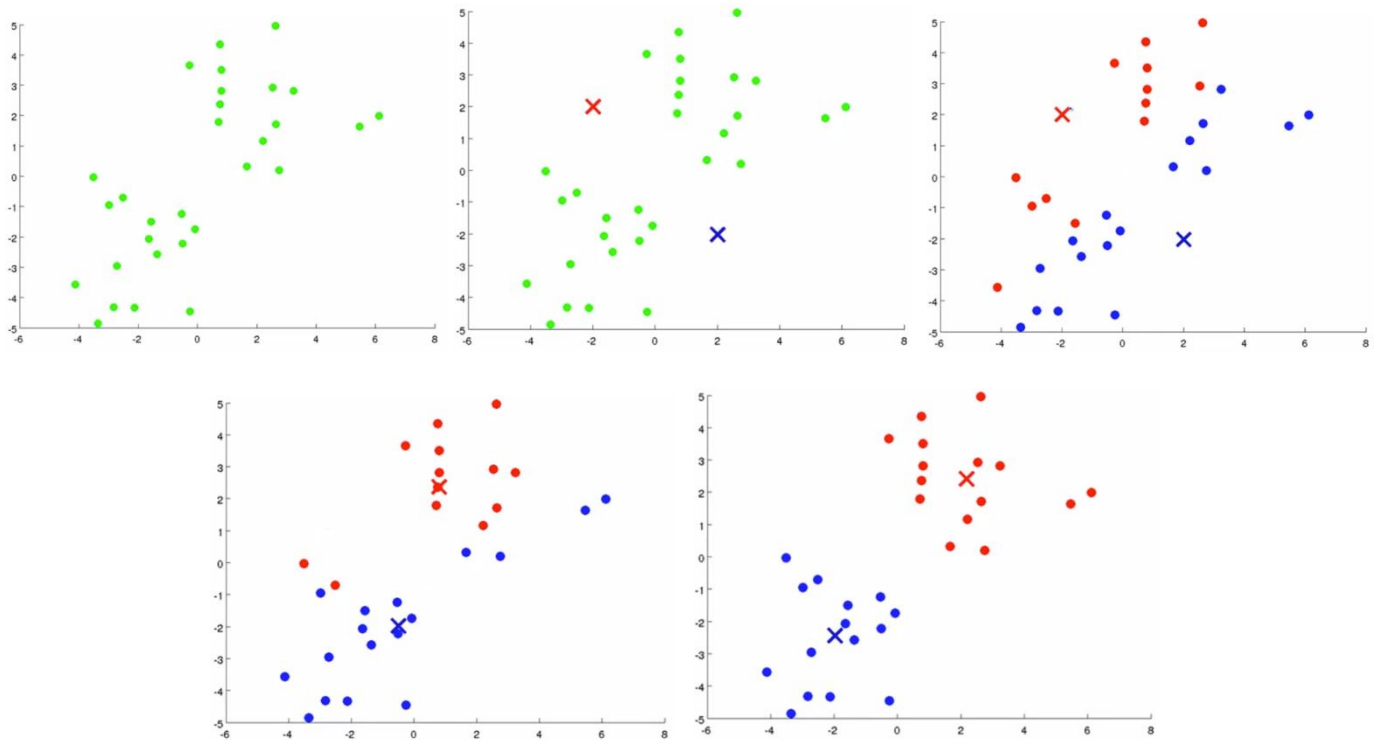
Objective Function

- The objective of K-Means is to minimize the sum of squared distances between data points and their assigned centroids.

$$J = \sum_{j=1}^K \sum_{i \in C_j} ||x^{(i)} - \mu_j||^2$$

- Where K is the number of clusters, C_j is the set of points in cluster j , $x^{(i)}$ is example i in cluster C_j , and μ_j is the centroid of cluster j .

■ **Example**
Let $K = 2$.



Practice Problem

1. Apply K-means clustering algorithm to the following training set having just one feature. Take $K=2$.
Training set: {2, 3, 4, 10, 11, 12, 20, 25, 30}

Randomly selecting $\mu_1 = 4$ and $\mu_2 = 12$. (4-2, 12-2)

Iteration	Centroids		Training Set								
	μ_1	μ_2	2	3	4	10	11	12	20	25	30
1	4	12	min(2,10)	min(1,9)	min(0,8)	min(6,2)	min(7,1)	min(8,0)	min(16,8)	min(21,13)	min(26,18)
	3	18	$c^{(1)}=1$	$c^{(2)}=1$	$c^{(3)}=1$	$c^{(4)}=2$	$c^{(5)}=2$	$c^{(6)}=2$	$c^{(7)}=2$	$c^{(8)}=2$	$c^{(9)}=2$
2	3	18	min(1,16)	min(0,15)	min(1,14)	min(7,8)	min(8,7)	min(9,6)	min(17,2)	min(22,7)	min(27,12)
	4.75	19.6	$c^{(1)}=1$	$c^{(2)}=1$	$c^{(3)}=1$	$c^{(4)}=1$	$c^{(5)}=2$	$c^{(6)}=2$	$c^{(7)}=2$	$c^{(8)}=2$	$c^{(9)}=2$
3	4.75	19.6	$c^{(1)}=1$	$c^{(2)}=1$	$c^{(3)}=1$	$c^{(4)}=1$	$c^{(5)}=1$	$c^{(6)}=1$	$c^{(7)}=2$	$c^{(8)}=2$	$c^{(9)}=2$
	7	25	$c^{(1)}=1$	$c^{(2)}=1$	$c^{(3)}=1$	$c^{(4)}=1$	$c^{(5)}=1$	$c^{(6)}=1$	$c^{(7)}=2$	$c^{(8)}=2$	$c^{(9)}=2$
4	7	25	$c^{(1)}=1$	$c^{(2)}=1$	$c^{(3)}=1$	$c^{(4)}=1$	$c^{(5)}=1$	$c^{(6)}=1$	$c^{(7)}=2$	$c^{(8)}=2$	$c^{(9)}=2$
	7	25	$c^{(1)}=1$	$c^{(2)}=1$	$c^{(3)}=1$	$c^{(4)}=1$	$c^{(5)}=1$	$c^{(6)}=1$	$c^{(7)}=2$	$c^{(8)}=2$	$c^{(9)}=2$

2. Consider the following dataset in which weight (in kg) and height (in cm) are given for n persons.

Training set: {(37,137), (60, 163),}

Here $m = 2$, hence x and μ are 2-dimensional vectors.

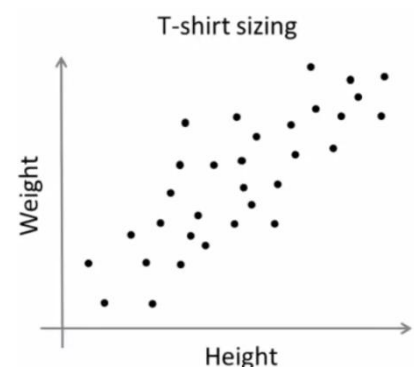
Let $\mu_1 = (47, 150)$ and $\mu_2 = (70, 180)$

Distance of $x^{(1)}$ from $\mu_1 = \sqrt{(37 - 47)^2 + (137 - 150)^2} = 16.4$

Distance of $x^{(1)}$ from $\mu_2 = \sqrt{(37 - 70)^2 + (137 - 180)^2} = 54.2$

Let cluster 1 $\mu_1 = (47, 150)$ contain both x_1 and x_2 , then new values for centroids can be calculated as:

$$\mu_1 = \left(\frac{37+60}{2}, \frac{137+163}{2} \right)$$

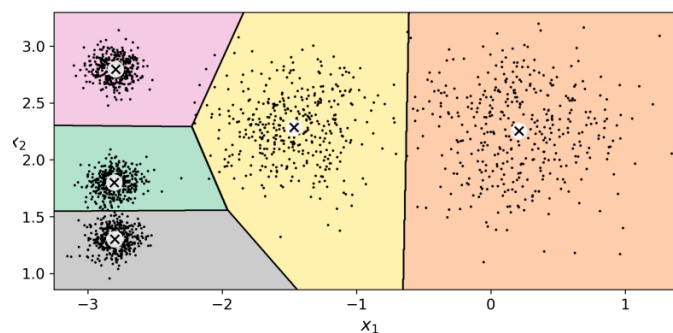


Issues with K-Means Clustering

- K-means is a fast and scalable clustering algorithm.
- The algorithm is guaranteed to converge in a finite number of iterations and will not oscillate indefinitely.
- This is because the objective (cost) function decreases or remains unchanged after every iteration.
- However, K-Means has several limitations and practical issues.

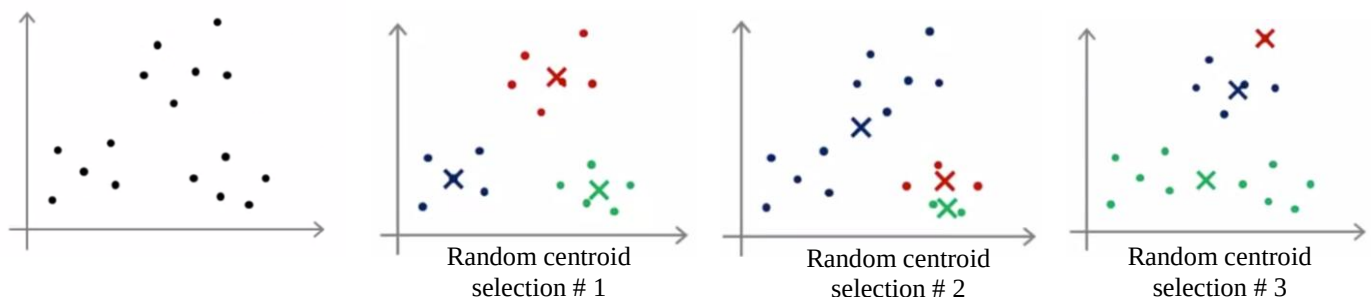
Sensitivity to Cluster Shape and Size

- K-Means performs well when clusters are compact and have similar sizes and densities.
- The algorithm does not behave well when clusters have:
 - very different diameters,
 - different densities,
 - non-spherical shapes,
 - overlapping regions.
- **Example:**
Since K-Means assigns points based only on distance to centroids, some points near cluster boundaries may be incorrectly assigned.



Sensitivity to Centroid Initialization

- Although K-Means is guaranteed to converge, it may not converge to the optimal solution.
- Random initialization of centroids can lead to poor clustering results.
- Different initial centroids may produce different final clusters.
- The algorithm may converge to a local minimum instead of the global minimum.
- **Example:**
The following set of figures show a dataset (leftmost) and three possible random assignments of centroids (rightmost), each one resulting in different clustering.



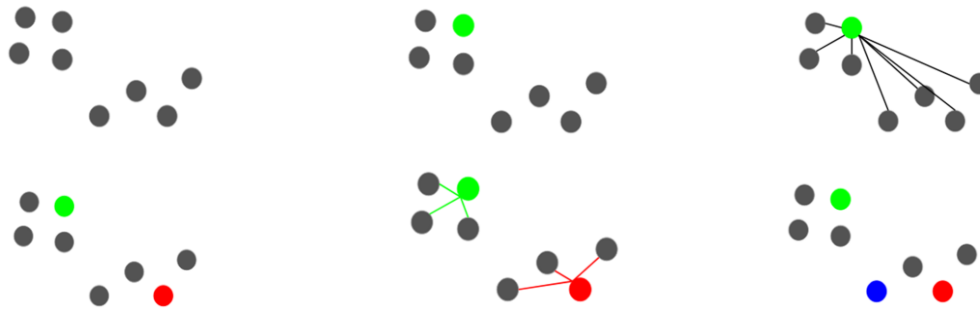
Initializing Centroids

Multiple Random Initializations

- One approach to reduce poor clustering due to random initialization is to run K-Means multiple times using different random centroids.
- Compute the cost function and select the clustering with the lowest cost.
- This approach works well when K is small (typically K=2 to 10).
- For larger values of K, multiple random initializations usually provide only slight improvements.

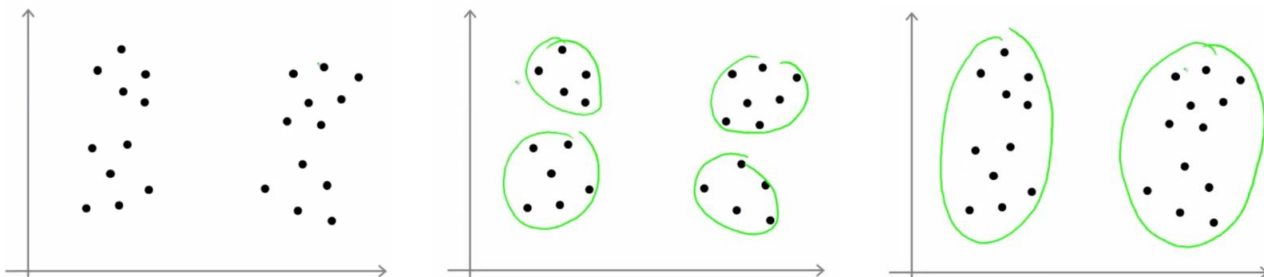
K-Means++ Initialization

- K-Means++ is an improved centroid initialization technique that helps produce better clustering results.
- Steps of K-Means++:
 - Choose the first centroid randomly from the training examples.
 - Compute the squared distance of every point from its nearest chosen centroid.
 - Select the next centroid with probability proportional to its squared distance from the nearest centroid.
 - Repeat steps 2 and 3 until K centroids are selected.
- Points that are farther away from existing centroids are more likely to be selected as new centroids.
- **Example**
Choosing 3 centroids for the following training data.



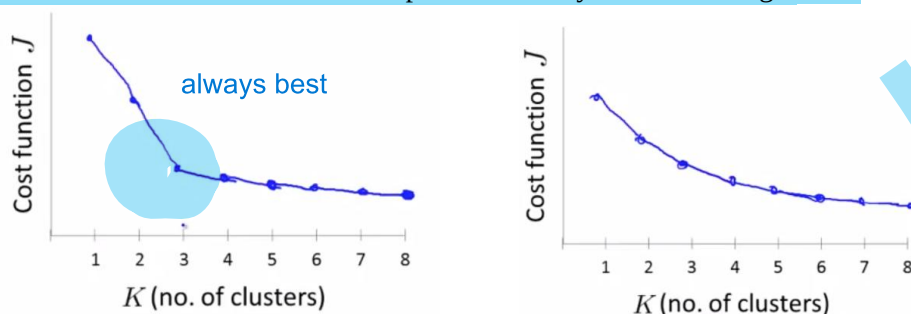
Choosing the Number of Clusters

- One of the main challenges in K-Means clustering is selecting the appropriate value of K.
- There is **no universal method for determining the optimal number of clusters**.
- In many datasets, it is ambiguous how many clusters actually exist.
- **Example:** Two possible clustering (rightmost two) for the dataset (leftmost).



Elbow Method

- A commonly used technique for selecting K is the elbow method.
- The elbow plot is a graph between number of clusters (K), and the cost function.
- As K increases, the distortion decreases.
- The optimal value of K is often chosen at the “elbow point,” where the decrease in distortion begins to slow down significantly.
- However, the elbow method does not always clearly identify the best value of K.
- **Example:** Elbow method works for the left example but not very well for the right one.



Choosing K Based on Downstream Tasks

- Sometimes the value of K is selected based on how useful the clustering is for a later task or application.
- The best clustering may depend on the final purpose of the analysis.
- Example

